10. The Functional Equations of the Isomorphic Mapping

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Following Dedekind,[1] by an *isomorphic mapping of the field* $\mathfrak{A}$ *onto a system* $\mathfrak{B}$ – which subsequently also proves to be a field – one understands *such a single-valued correspondence that to every element of* $\mathfrak{A}$ *there corresponds one and only one element of* $\mathfrak{B}$; *and that to the sum, the difference, the product, and the quotient of any two elements of* $\mathfrak{A}$ *there are again assigned the sum, difference, product, and quotient of the uniquely corresponding elements of* $\mathfrak{B}$.

Let such a mapping be mediated by a – by the preceding, single-valued – function $f(z)$. If z then runs through all elements $x, y, \ldots$ of the field $\mathfrak{A}$, then $f(z)$ is characterized by the functional equations

$$(1)\quad f(x+y) = f(x) + f(y); \quad (2)\quad f(x-y) = f(x) - f(y);$$
$$(3)\quad f(x \cdot y) = f(x) \cdot f(y); \qquad (4)\quad f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}.$$

In what follows the most general single-valued solutions $f(z)$ *of these functional equations are to be given when* $x, y, \ldots$ *run through all real and complex numbers; thus, the most general function* $f(z)$ *which effects an isomorphic mapping of the field of all complex numbers.*[2] The solution for arbitrary fields follows quite analogously.

G. Hamel[3] constructed the most general solutions of the functional equation (1) by means of a *linear* basis of all numbers. The construction given here for the solutions of (1) to (4) – hence of the mapping function relative to which the rational and algebraic operations are invariant – uses the *rational* and *algebraic* basis of all numbers. After the isomorphic mapping is discussed more closely in 1, necessary conditions for $f(z)$ are given in 2; in 3 these are recognized as sufficient and lead to the actual construction.[4] – The solutions of (1) to (4), with the known exceptions $f(z) = z$ or $\overline{z}$, are discontinuous; in 4 it is further shown that they are "extremely discontinuous" – a fact which G. Hamel proved for the real solutions of (1), but which need not be fulfilled there in the complex domain.[5]

1. We first draw some consequences from the functional equations (1) to (4), directly for general fields $\mathfrak{A}$. By (1), single-valuedness implies $f(0) = 0$. If $f(y)$ does not vanish, then the original y cannot vanish either, and the functional equations (1) to (4) therefore show that *the mapping system* $\mathfrak{B}$ *of all values* $f(z)$ *again forms a field.* Conversely, if the initial condition $f(0) = 0$ is prescribed, then (2) implies the single-valuedness of the function $f(z)$: *the requirement of single-valuedness and the initial condition* $f(0) = 0$ *are therefore equivalent.* From (4) it follows that $f(z)$ cannot vanish identically; from (3) further, that $f(z)$ vanishes only for $z = 0$. For from $f(y) = 0$ and $y \neq 0$ it would follow, since z/y also belongs to $\mathfrak{A}$, that

$$f(z) = f\left(\frac{z}{y} \cdot y\right) = f\left(\frac{z}{y}\right) \cdot f(y) = 0,$$

which contradicts (4) by making $f(z)$ vanish identically. From $f(x) = f(y)$ it follows therefore that $x = y$; that is, to each value of $f(z)$ there also corresponds one and only one value of z: $f(z)$ *is a reversibly single-valued function of* z. As the functional equations show, the mapping mediated by the inverse function is again isomorphic. Since every rational operation is composed of a finite number of additions, multiplications, and their inverses, one can also say: *By the isomorphic mapping a one-to-one relation is established between the elements of* $\mathfrak{A}$ *and of* $\mathfrak{B}$, *in such a way that all rational relations holding among finitely many elements of* $\mathfrak{A}$,

$$F(x, y, \ldots, t) = 0,$$

[1] Cf. for example Dirichlet-Dedekind, *Lectures on Number Theory* (4th ed.), Supplement XI, § 161. The designation "isomorphic mapping" or "isomorphism", modeled on group theory, occurs only in the later literature; Dedekind speaks of the "permutations of a field".

[2] This question was occasionally put to me by Mr. Landau. As I noticed afterward, part of the solutions already occurs in H. Lebesgue, Sur les transformations ponctuelles, transformant les plans en plans, Atti Acad. Torino, 1906/07; and likewise implicitly in A. Ostrowski, Über einige Fragen der allgemeinen Körpertheorie, Crelle's Journal 143 (1913), § 2.

[3] G. Hamel, Eine Basis aller Zahlen und die unstetigen Lösungen der Funktionalgleichung: $f(x+y) = f(x) + f(y)$, Math. Ann. 60, p. 459 (1905).

[4] Cf. the parallel considerations in my paper: Die allgemeinsten Bereiche aus ganzen transzendenten Zahlen. Math. Ann. 77, p. 103 (1915), especially §§ 8 and 10.

[5] Hamel's term "totally discontinuous" is used in a different sense in the rest of the literature.

are preserved in $\mathfrak{B}$, and conversely.

It should also be noted that for a field $f(z)$ is already characterized by the functional equations (1) and (3), by the requirement that $f(z)$ not vanish identically, and by the initial condition $f(0) = 0$. Indeed, (2) follows from (1) by replacing x by the element $(x-y)$ belonging to $\mathfrak{A}$; then (2) and $f(0) = 0$ give the single-valuedness of $f(z)$. As was shown above, it follows further from (3), by replacing x by the element x/y belonging to $\mathfrak{A}$, that $f(y)$ cannot vanish for $y \neq 0$, and hence the validity of (4) and at the same time one-to-oneness. If instead of a field one takes an integral domain $\mathfrak{S}$ as basis, x/y need not belong to $\mathfrak{S}$, and from (3) one cannot infer one-to-oneness. Here, however, the following most familiar formulation suffices: an isomorphic mapping is a *one-to-one* correspondence between the elements of $\mathfrak{S}$ and $\mathfrak{S}'$ such that to the sum and the product there always correspond sum and product.[6]

2. From now on, for simplicity, instead of $\mathfrak{A}$ let the field $\mathfrak{C}$ of all real and complex numbers be taken as the basis. The first matter is to discuss the systems of values which $f(z)$ assumes. From (3) or (4) it follows that $f(1) = 1$, and thence, by the functional equations, for every *rational number* α, $f(\alpha) = \alpha$; *therefore every rational number corresponds to itself under the mapping.* Since an *algebraic number* β satisfies an irreducible equation with rational numerical coefficients $F(\beta, \alpha) = 0$, no. 1 and the preceding observation give, for $f(\beta)$, the equation $F(f(\beta), \alpha) = 0$; *every algebraic number therefore goes into itself or into one of its conjugates*, and by one-to-oneness the totality of algebraic numbers again corresponds to the field of all algebraic numbers. In order to recognize the behavior of the *transcendental* numbers, and at the same time to separate the conjugate values, we take as basis a *rational basis of all numbers*,[7] i.e. a system Θ of numbers ϑ which permits all numbers to be expressed rationally – with rational numerical coefficients[8] – while in at least one well-ordering no basis number is rationally expressible through finitely many preceding ones. The existence of this basis follows exactly analogously to linear and algebraic bases from the well-ordering theorem.

Let $\mathfrak{C}$ be subjected to a well-ordering Ω. Then every number is either expressible rationally through finitely many preceding numbers, or is rationally independent of the preceding ones. These latter numbers form the basis Θ, and induction shows that indeed every number is rationally expressible through finitely many ϑ's.

For each basis number ϑ, the "segment field $\mathfrak{K}(\vartheta)$" is defined: it consists of all rational combinations of those basis numbers which precede ϑ in the well-ordering. As the segment field of the first basis number one is to regard the field $\mathfrak{R}$ of all rational numbers. The number ϑ can display two sorts of behavior with respect to $\mathfrak{K}(\vartheta)$: it can be algebraically dependent on $\mathfrak{K}(\vartheta)$, or algebraically independent of it. Since, by no. 1, all relations valid in $\mathfrak{C}$ are also valid in the mapping field, and conversely, the totality of values $f(\vartheta)$ corresponding to the ϑ's forms a basis of the mapping range, well-ordered by means of the well-ordering of Θ itself. In particular, the segment field $\mathfrak{K}(\vartheta)$ corresponds to the segment field $\mathfrak{K}(f(\vartheta))$; and according as ϑ is algebraically dependent on, or independent of, $\mathfrak{K}(\vartheta)$, the same holds for $f(\vartheta)$ with respect to $\mathfrak{K}(f(\vartheta))$. In the dependent case the irreducible equations for ϑ and $f(\vartheta)$ also correspond. The totality of those values ϑ which are in each case algebraically independent of $\mathfrak{K}(\vartheta)$ form – as induction again shows – an *algebraic basis of all numbers*,[9] i.e. a system H of numbers η which permits all numbers to be expressed algebraically, while no algebraic relations hold among the η's themselves. To this algebraic basis H there corresponds in the mapping range again an algebraic basis Z, which by one-to-oneness has the same cardinality as H, namely the cardinality of the continuum, since algebraic extension is known not to increase cardinality.[10] From the knowledge of the values $f(\vartheta)$, however, $f(z)$ is immediately obtained for all systems of values, since from $z = \psi(\vartheta)$ there always follows $f(z) = \psi(f(\vartheta))$.

3. We now show that the necessary conditions found above actually also provide the construction of $f(z)$. The basis Z corresponding one-to-one to the algebraic basis H is constructed and assigned to H as follows. Put a second well-ordering Ω' in place, which supplies an algebraic basis Z' of all numbers. Let Z, with elements ξ, be a subset of Z' of the same cardinality as Z', and hence as H. To each element η_i of H assign one and only one element ξ_i, with the same numbering, by the rule: ξ_i is the *first* of all elements of Z in the well-ordering Ω' which has not been assigned to any element of H preceding η_i.

Now define $f(z)$ as follows:

(a) *for all rational numbers α, by $f(0) = 0$, $f(\alpha) = \alpha$;*

[6] Cf. on no. 1: Dedekind, loc. cit., § 161.

[7] Cf. my cited paper on "whole transcendental numbers", § 6.

[8] By a "rational function" I shall always mean one whose coefficients are also rational numbers.

[9] Cf. E. Zermelo, Über ganze transzendente Zahlen, § 1, Math. Ann. 75 (1914).

[10] Cf. E. Steinitz, Algebraische Theorie der Körper, Crelle's Journal 137 (1910), § 24.

(b) *for all quantities of the algebraic basis* H, *by* $f(\eta_i) = \xi_i$;

(c) *for all remaining quantities* ϑ *of the rational basis* Θ – *those algebraically dependent on the segment field* $\mathfrak{K}(\vartheta)$, *and hence satisfying an equation* $F(\vartheta; \vartheta_{i_1}, \ldots, \vartheta_{i_\nu}) = 0$ *irreducible in* $\mathfrak{K}(\vartheta)$ – *as a root of the equation* $F(f(\vartheta); f(\vartheta_{i_1}), \ldots, f(\vartheta_{i_\nu})) = 0$;

(d) *for all remaining values* z, *which by the definition of* Θ *are representable as* $z = \psi(\vartheta_{k_1}, \ldots, \vartheta_{k_\nu})$, *by* $f(z) = \psi(f(\vartheta_{k_1}), \ldots, f(\vartheta_{k_\nu}))$.[11]

To show that the so-defined $f(z)$ actually satisfies the functional equations (1) to (4), by no. 1 it suffices to prove this for (1) and (3), since $\mathfrak{C}$ is a field and $f(z)$, because of (a) and (b), cannot vanish identically and also satisfies the initial condition $f(0) = 0$. By means of the rational basis Θ, express $x, y, x + y$ as

$$x = \psi_1(\vartheta_1 \cdots \vartheta_t), \quad y = \psi_2(\vartheta_1 \cdots \vartheta_t), \quad (x + y) = \psi_3(\vartheta_1 \cdots \vartheta_t).$$

Thus the proof that $f(z)$ satisfies the functional equation (1),

$$f(x) + f(y) - f(x + y) = 0,$$

is, by (d), identical with showing that from

$$\psi_1(\vartheta) + \psi_2(\vartheta) - \psi_3(\vartheta) = \frac{H_1(\vartheta)}{H_2(\vartheta)} = 0$$

there always follows

$$\psi_1(f(\vartheta)) + \psi_2(f(\vartheta)) - \psi_3(f(\vartheta)) = \frac{H_1(f(\vartheta))}{H_2(f(\vartheta))} = 0,$$

where H_1, H_2 denote integral rational functions of their arguments. The functional equation (3) leads to precisely the same form of condition; hence the satisfaction of all the functional equations is identical with the following condition – necessary also by nos. 1 and 2: for *every* integral rational function $H(\vartheta)$, from $H(\vartheta) = 0$ there always follows $H(f(\vartheta)) = 0$, and from $H(\vartheta) \neq 0$ there follows $H(f(\vartheta)) \neq 0$, the latter because of the occurrence of the denominator.

That this is indeed the case is shown by induction. Suppose that in $H(\vartheta_1 \cdots \vartheta_t)$ the basis quantity ϑ_t is the last, in the well-ordering, among $\vartheta_1 \cdots \vartheta_t$.[12] Then $H(\vartheta_1 \cdots \vartheta_t) = 0$ may be regarded as a relation satisfied by ϑ_t with respect to the segment field $\mathfrak{K}(\vartheta_t)$, and likewise $H(f(\vartheta_1) \cdots f(\vartheta_t)) = 0$ as a relation satisfied by $f(\vartheta_t)$ with respect to $\mathfrak{K}(f(\vartheta_t))$. If not all relations $H(\vartheta) = 0$ were also fulfilled by $f(\vartheta)$, and conversely, then there would have to be a first basis quantity, say ϑ_0, for which at least one relation valid with respect to $\mathfrak{K}(\vartheta_0)$ failed to be preserved in the mapping field, or conversely, while the preceding segment fields $\mathfrak{K}(\vartheta_0)$ and $\mathfrak{K}(f(\vartheta_0))$ were isomorphic. In particular, by (a), the segment field of the first basis quantity of Θ, the field $\mathfrak{R}$ of all rational numbers, is isomorphic to its mapping field. Suppose now that $H(\vartheta_0)$ has the form

$$H(\vartheta_0) = a_0(\vartheta) \cdot \vartheta_0^\chi + a_1(\vartheta) \cdot \vartheta_0^{\chi-1} + \cdots + a_\chi(\vartheta),$$

where the $a_i(\vartheta)$ are quantities from $\mathfrak{K}(\vartheta_0)$. Then there are two possibilities.

1) ϑ_0 is algebraically independent of $\mathfrak{K}(\vartheta_0)$. Then $H(\vartheta_0) = 0$ is fulfilled identically in ϑ_0,[13] so that $a_i(\vartheta) = 0$ for all coefficients, and hence by the isomorphism of $\mathfrak{K}(\vartheta_0)$ and $\mathfrak{K}(f(\vartheta_0))$ also $a_i(f(\vartheta)) = 0$. From $H(\vartheta_0) = 0$ there follows, therefore, always $H(f(\vartheta_0)) = 0$. Conversely, suppose $H(f(\vartheta_0)) = 0$. Since ϑ_0, being algebraically independent of $\mathfrak{K}(\vartheta_0)$, belongs to the algebraic basis H, $f(\vartheta_0)$ belongs by (b) to the algebraic basis Z and is algebraically independent of $\mathfrak{K}(f(\vartheta_0))$. Thus all $a_i(f(\vartheta))$ vanish, and by the isomorphism all $a_i(\vartheta)$ vanish. From $H(f(\vartheta_0)) = 0$ follows $H(\vartheta_0) = 0$, or from $H(\vartheta_0) \neq 0$ also $H(f(\vartheta_0)) \neq 0$.

2) ϑ_0 is algebraically dependent on $\mathfrak{K}(\vartheta_0)$. Then $H(\vartheta_0)$ is divisible by the equation $F(\vartheta_0)$ irreducible with respect to $\mathfrak{K}(\vartheta_0)$; that is, identically in t,

$$H(t; a_i(\vartheta)) = F(t; \vartheta) \cdot G(t; \vartheta_i).$$

Since all the coefficients which occur belong to $\mathfrak{K}(\vartheta_0)$, the corresponding relation is fulfilled in the isomorphic field $\mathfrak{K}(f(\vartheta_0))$:

$$H(t; a_i(f(\vartheta))) = F(t; f(\vartheta)) \cdot G(t; f(\vartheta)).$$

[11] In (d), (a) is contained as a special case.

[12] The expressions $H(\vartheta)$ which are of degree zero in the ϑ's go over into themselves under the mapping and therefore need not be investigated further.

[13] Because $x, y, \ldots$ can be expressed by the ϑ's, this form of relation can very well occur.

But by (c), $f(\vartheta_0)$ was chosen to be a root of $F(t; f(\vartheta))$. Hence from $H(\vartheta_0) = 0$ there follows $H(f(\vartheta_0)) = 0$.

Conversely, if $H(f(\vartheta_0)) = 0$, then the divisibility of $H(t; a_i(f(\vartheta)))$ by the function $F(t; f(\vartheta))$, irreducible with respect to $\mathfrak{K}(f(\vartheta_0))$, implies for the isomorphic field $\mathfrak{K}(\vartheta_0)$ the divisibility of $H(t; a_i(\vartheta))$ by $F(t; \vartheta)$. And since $F(t; f(\vartheta))$ arose from the irreducible equation for ϑ_0 by the isomorphic relation, and this relation is one-to-one, $F(t; \vartheta)$ necessarily represents the irreducible function whose root is ϑ_0. Thus from $H(f(\vartheta_0)) = 0$ follows $H(\vartheta_0) = 0$, or from $H(\vartheta_0) \neq 0$ also $H(f(\vartheta_0)) \neq 0$.

From isomorphic segment fields $\mathfrak{K}(\vartheta_0)$ and $\mathfrak{K}(f(\vartheta_0))$, therefore, isomorphic fields also arise by adjunction of ϑ_0 and $f(\vartheta_0)$, respectively; the assumption that this fails for some ϑ_0 leads to a contradiction. It is thus proved that *the function $f(z)$ defined by (a) to (d) actually represents a solution of the functional equations (1) to (4), and, by 2, the most general one.*

All considerations used in constructing $f(z)$ have used only the fact that the totality $\mathfrak{C}$ of all numbers forms a field; hence they hold for every abstractly defined field $\mathfrak{A}$. Here one should note that the field $\mathfrak{R}$ of rational numbers is replaced by the "prime field" of $\mathfrak{A}$, i.e. by the field derived from the unit element of $\mathfrak{A}$ by the operations of addition and multiplication and their inverses. If, therefore, in (a) to (d) one replaces the rational numbers by the elements of the prime field, then *the function $f(z)$ so obtained effects the most general isomorphic mapping of an arbitrary abstractly defined field.*

4. It remains to show that the functions $f(z)$ defined for the field $\mathfrak{C}$ of all real and complex numbers – which, except for $f(z) = z$ or $\overline{z}$, are known to be discontinuous – are *extremely discontinuous*; that is, in every neighborhood of any real or complex number z_0 there are values z for which $f(z)$ comes arbitrarily close to any prescribed value Z_0. This extreme discontinuity belongs, as G. Hamel proved loc. cit., to the *real* solutions $\varphi(x)$, defined for all *real* numbers, of the functional equation (1). But, as the examples below show, it need not belong to the solutions defined on the complex numbers. Hamel argues as follows. If $\varphi(x)$ differs from $C \cdot x$, then there are at least two values of the linear basis, say x_1 and x_2, such that $\varphi(x_1) : x_1 \neq \varphi(x_2) : x_2$. Then the two equations

$$\alpha x_1 + \beta x_2 = a; \qquad \alpha\varphi(x_1) + \beta\varphi(x_2) = b$$

have a solution in real numbers α, β; consequently a and b can be approximated arbitrarily closely by rational numbers α, β. Because α, β are rational, $\alpha\varphi(x_1) + \beta\varphi(x_2)$ is then actually $\varphi(\alpha x_1 + \beta x_2)$. This argument breaks down for complex systems of values. In fact one can give, also in the complex domain, solutions of (1) which are discontinuous but not extremely discontinuous; for example,

$$\varphi(z) = \varphi(x + iy) = \varphi(x) + i(y + \varphi(x)),$$

where $\varphi(x)$ denotes a real solution which is extremely discontinuous in the real domain; or, still more simply, $\varphi(z) = \varphi(x)$. In both cases the real part of $\varphi(z)$ is extremely discontinuous, while the imaginary part is determined by the real part.

This behavior, and at the same time the behavior of the function $f(z)$, becomes completely clear if one introduces the concept of rank. We put $z = x + iy$, $\varphi(z) = X + iY$, and call $\varphi(z)$ of rank four, three, two, or one, respectively, according as there exist no, one, two, or three linearly independent relations

$$c_1 x + c_2 y + c_3 X + c_4 Y = 0,$$

where of course the c's are real numbers.

1) Let $\varphi(z)$ have *rank four.* Then there exist at least four values of the linear basis, say z_1, z_2, z_3, z_4, such that the determinant

$$\begin{vmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \\ X_1 & X_2 & X_3 & X_4 \\ Y_1 & Y_2 & Y_3 & Y_4 \end{vmatrix}$$

does not vanish. The equations

$$\begin{aligned} \alpha x_1 + \beta x_2 + \gamma x_3 + \delta x_4 &= a, \\ \alpha y_1 + \cdots + \delta y_4 &= b, \\ \alpha X_1 + \cdots + \delta X_4 &= A, \\ \alpha Y_1 + \cdots + \delta Y_4 &= B \end{aligned}$$

therefore have a solution in real numbers $\alpha, \ldots, \delta$, and a, b, A, B can be approximated arbitrarily closely by rational numbers $\alpha, \beta, \gamma, \delta$. Moreover one has

$$\alpha(X_1 + iY_1) + \beta(X_2 + iY_2) + \gamma(X_3 + iY_3) + \delta(X_4 + iY_4) = \varphi(\alpha z_1 + \beta z_2 + \gamma z_3 + \delta z_4).$$

"In general", therefore, the solutions $\varphi(z)$ are extremely discontinuous also in the complex domain; the *discontinuity values* of $\varphi(z)$ in the neighborhood of $z_0 = a + ib$ *form a two-dimensional linear manifold.*

2) Let $\varphi(z)$ have *rank three.* Then there are at least three systems of values of the linear basis, say z_1, z_2, z_3, such that the above determinant has rank three. The values a, b, A, B satisfying the relation

$$c_1 a + c_2 b + c_3 A + c_4 B = 0,$$

and only these, can be approximated arbitrarily closely; the *discontinuity values* form a *one-dimensional linear manifold* determined by $z_0 = a + ib$. As the examples given show, this case can actually occur.

3) Let $\varphi(z)$ have *rank two.* Since one can always choose two complex numbers z_1, z_2 such that

$$\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} \neq 0,$$

the relations have the form

$$X = \lambda_1 x + \lambda_2 y; \qquad Y = \mu_1 x + \mu_2 y.$$

The resulting expression

$$\varphi(z) = (\lambda_1 x + \lambda_2 y) + i(\mu_1 x + \mu_2 y)$$

is precisely the most general *continuous* solution of the functional equation (1) in the domain of complex numbers.

4) Let $\varphi(z)$ have *rank one.* This case cannot occur in the domain of complex numbers; in the domain of real numbers it means the *continuous* solution $\varphi(x) = C \cdot x$.

We now show that the *solutions $f(z)$ of the functional equations (1) to (4) defined for all real and complex numbers can have only rank two or rank four.* Here rank two gives only the two *continuous* functions $f(z) = z$ or $\overline{z}$, as the specialization $f(1) = 1$ and $f(i) = \pm i$, combined with 3), shows. Since rank one has already been excluded, it remains to exclude rank three. We therefore assume that at least one relation exists,

$$c_1 x + c_2 y + c_3 X + c_4 Y = 0,$$

so that $f(z)$ has rank three, or possibly lower rank, and show that under this assumption the latter case always occurs. As again the specialization $f(1) = 1$, $f(i) = \pm i$ shows, the relation becomes

$$c_3(X - x) + c_4(Y \mp y) = 0,$$

where c_3 and c_4 do not both vanish. First suppose, say, $c_4 = 0$. The relation $(X - x) = 0$ then means that to a purely imaginary z there also corresponds a purely imaginary $f(z)$. Thus for every real y one has $f(iy) = i \cdot \mathfrak{Y}$, where $\mathfrak{Y}$ is again real. From this, because

$$f(iy) = \pm i f(y),$$

one obtains also $f(y) = \pm\mathfrak{Y}$; hence to every real z there also corresponds a real $f(z)$, and because of $(X - x)$ all real values go into themselves. Thus one has only $f(z) = z$ or $\overline{z}$, so actually rank two. The conclusion is entirely analogous when $c_3 = 0$, $c_4 \neq 0$.

Now let c_3 and c_4 be different from zero, so that the relation can be written in the form

$$(Y \mp y) = c \cdot (X - x)$$

with real, nonzero, finite c. But this means that for $y = \pm cx$ one also has $Y = c \cdot X$. Hence, taking account of the functional equations, for every real x one has

$$f\{x \cdot (1 \pm ic)\} = \mathfrak{X}(1 + ic) = f(x) \cdot (1 + i f(c)),$$

where $\mathfrak{X}$ is again real. But here, by no. 1, the factor of $f(x)$ cannot vanish identically, and division gives

$$f(x) = \mathfrak{X}(\sigma + i\tau)$$

with real σ and τ, independent of x and $\mathfrak{X}$. In particular, to $x = 1$ there also belongs a real X_0, and the condition $1 = X_0(\sigma + i\tau)$ shows that necessarily $\tau = 0$, so that $f(x) = \sigma \cdot \mathfrak{X}$. Thus to every real z there also corresponds a real $f(z)$; equivalently, from $y = 0$ there necessarily follows $Y = 0$. Hence the linear relation for real z becomes $X - x = 0$;[14] every real value corresponds to itself. One again has only $f(z) = z$ or $\overline{z}$, so actually rank two. Rank three is thereby excluded in all cases. Since, however – as the investigations of the first sections show – discontinuous solutions actually exist, and these must therefore necessarily have rank four, the following theorem holds by 1): *"All solutions $f(z)$ of the functional equations (1) to (4) – with the exception of the continuous solutions $f(z) = z$ or $\overline{z}$ – are extremely discontinuous in the real and in the complex domain."*[15]

Erlangen, 30 October 1915.

[14] Here one uses the fact that $c \neq 0$, so that neither c_3 nor c_4 vanishes, whereas above only $c_4 \neq 0$ was used; therefore the case where one of these two vanishes had to be dealt with in advance.

[15] Lebesgue shows loc. cit. that the real and imaginary parts of $f(z)$ are both "non-measurable" functions of x and y; as the example at the beginning of no. 4, $\varphi(z) = \varphi(x) + i(y + \varphi(x))$, shows, this does not yet imply extreme discontinuity in the complex domain.